

CODE-LAYOUT, READABILITY AND REUSABILITY

Date	29 September 2026
Team ID	SWTID-2026-4636
Project Name	AI BlogNest API
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability

The AI BlogNest API follows a structured and modular code organization. Meaningful names, consistent formatting, reusable modules, validation, and error handling help make the backend easier to maintain, test, and extend.

S.No	Code Quality Parameter	Description	Followed
1	Consistent Indentation	Uniform spacing and indentation are used throughout the code.	Yes
2	Proper File Structure	Backend files and folders are logically organized into routes, controllers, models, middleware, and configuration.	Yes
3	Meaningful Variable Names	Variables and parameters clearly describe the data or purpose they represent.	Yes
4	Function / Method Names	Functions use descriptive names that indicate their purpose.	Yes
5	Code Comments	Important logic and API-related sections are explained with comments where required.	Yes
6	Modular Design	Authentication, blog operations, AI features, database access, and middleware are separated into reusable modules.	Yes
7	No Redundant Code	Repeated logic is minimized and common functionality is reused through modules and middleware.	Yes
8	Error Handling	API errors, validation failures, authentication errors, and database errors are handled with appropriate responses.	Yes

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Authentication Middleware	Node.js / Express.js / JWT	Protected blog and user API routes	High
2	User Model	MongoDB / Mongoose	Registration, login, and user-related operations	High
3	Blog Model & CRUD Module	Node.js / Express.js / MongoDB	Create, view, update, and delete blog posts	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
4	AI Service Module	Node.js / AI API	Blog ideas, content generation, summarization, and improvement	Medium
5	Error Handling / Validation	Express.js	API validation and common error responses	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The backend is organized into logical modules and folders.
Readability	5	Clear naming, consistent formatting, and simple module structure improve readability.
Reusability	5	Authentication, validation, database models, and common API logic are designed for reuse.
Documentation / Comments	4	Important modules and logic are documented where needed.
Overall Score	5	The codebase is structured, readable, modular, and suitable for future enhancement.